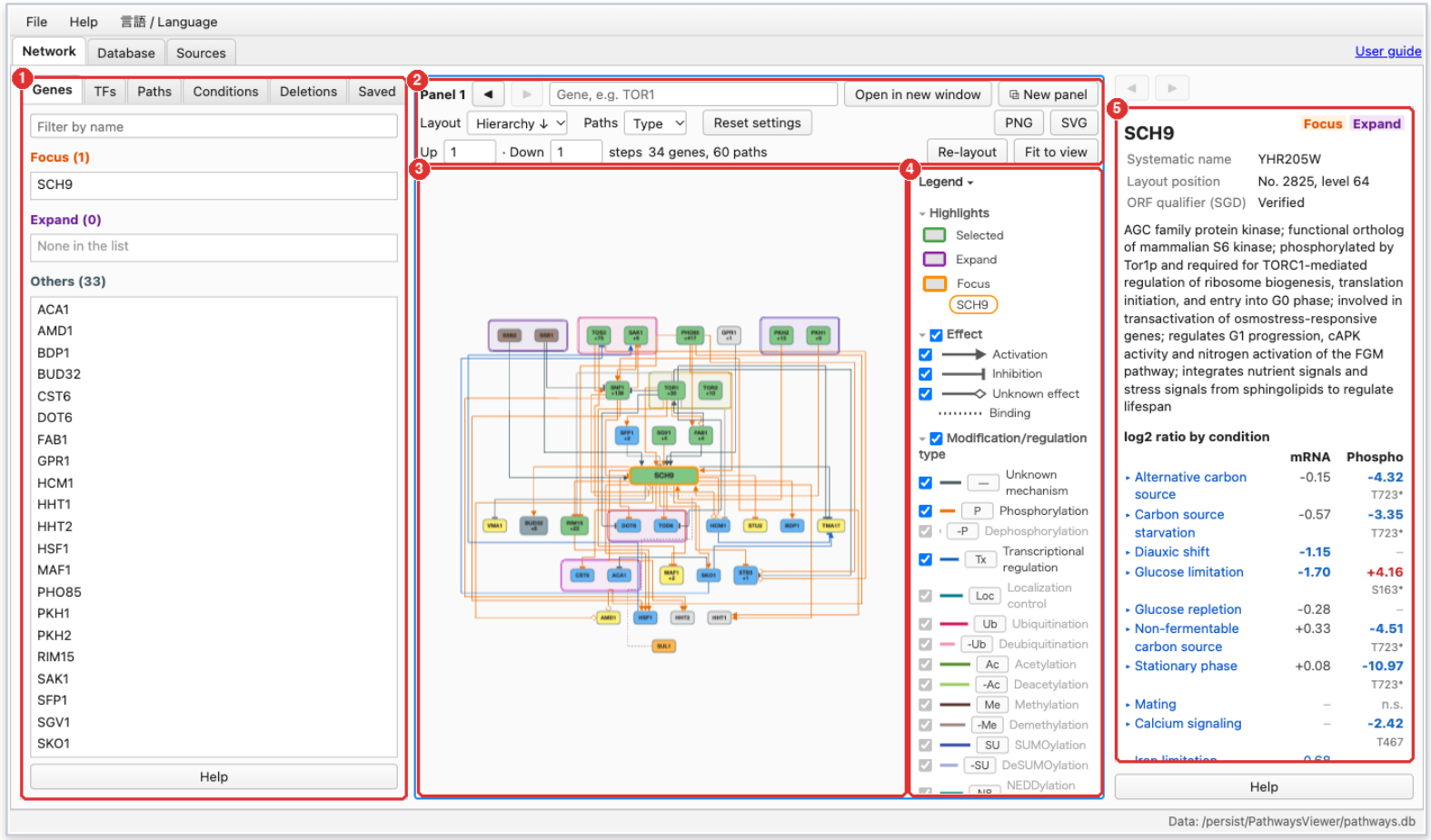
Pathways viewer for *Saccharomyces cerevisiae*

https://raico-lab.github.io/pathways_viewer_sc/

Runs in a web browser without installation. Open source under GPLv3.
A desktop browser such as Chrome, Edge, Safari or Firefox is recommended.
Source code: github.com/Raico-lab/pathways_viewer_sc

Interface



1 **Left-hand tabs**

Genes, transcription factors, paths, conditions, deletion strains and saved views

2 **Search and settings**

Gene search, layout, edge colouring and search depth

3 **Map**

Genes as nodes and relations as edges

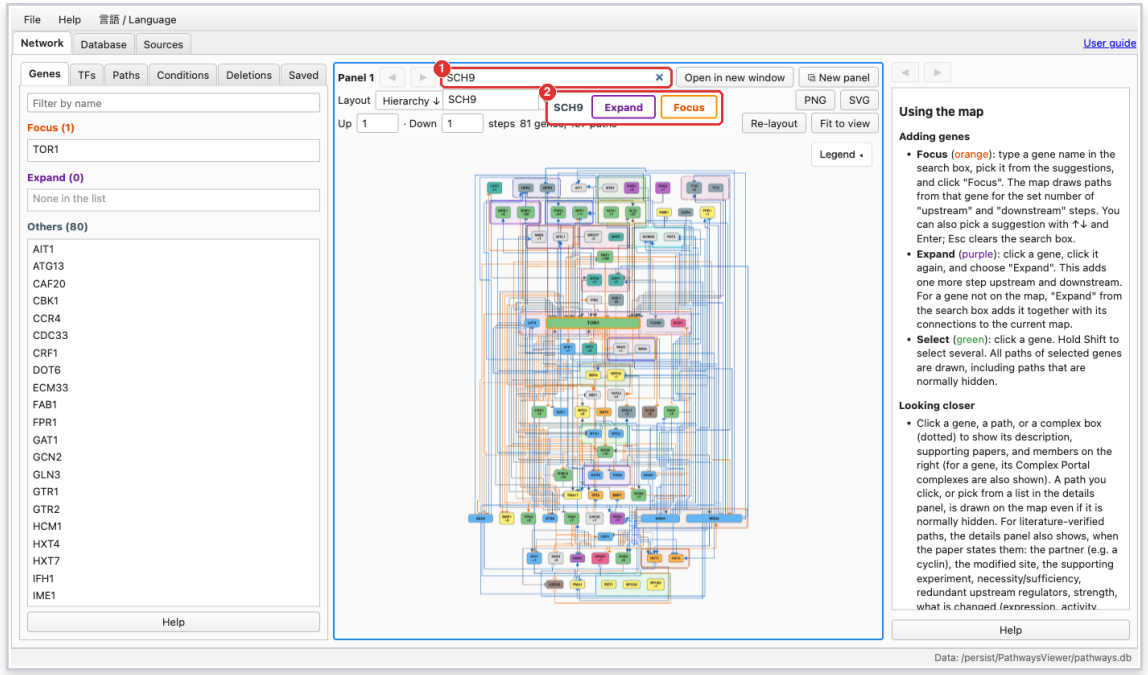
4 **Legend**

Symbol definitions; unticking an item hides that category

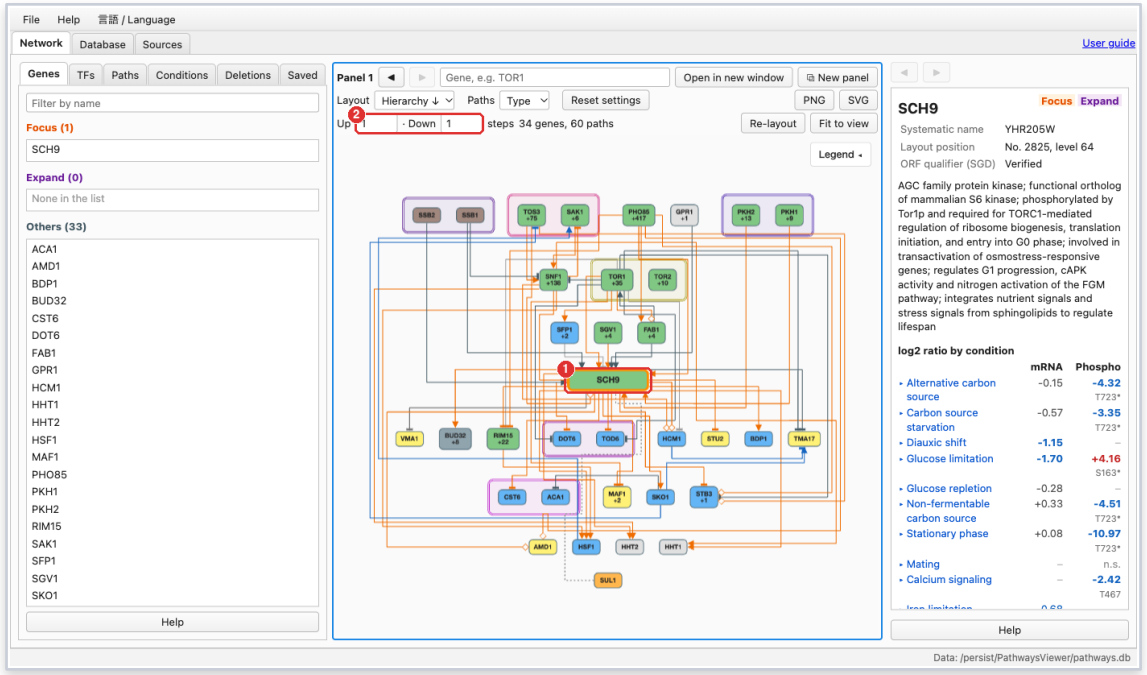
5 **Details**

Annotation, evidence and references for the selected item

Displaying a gene



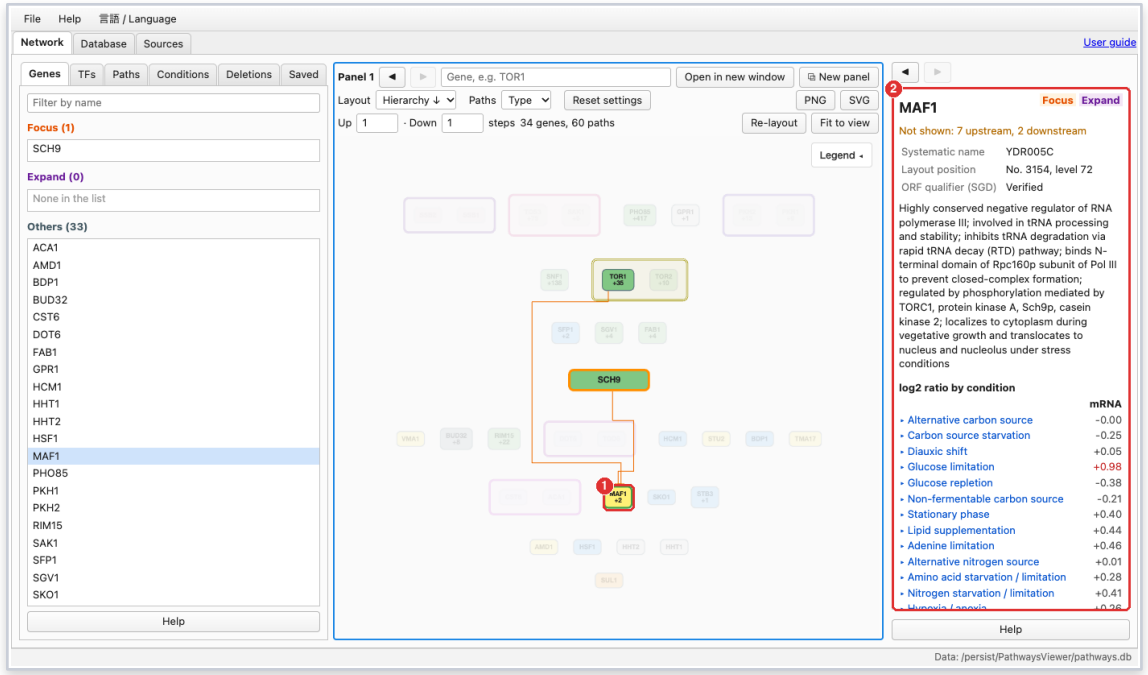
Gene search



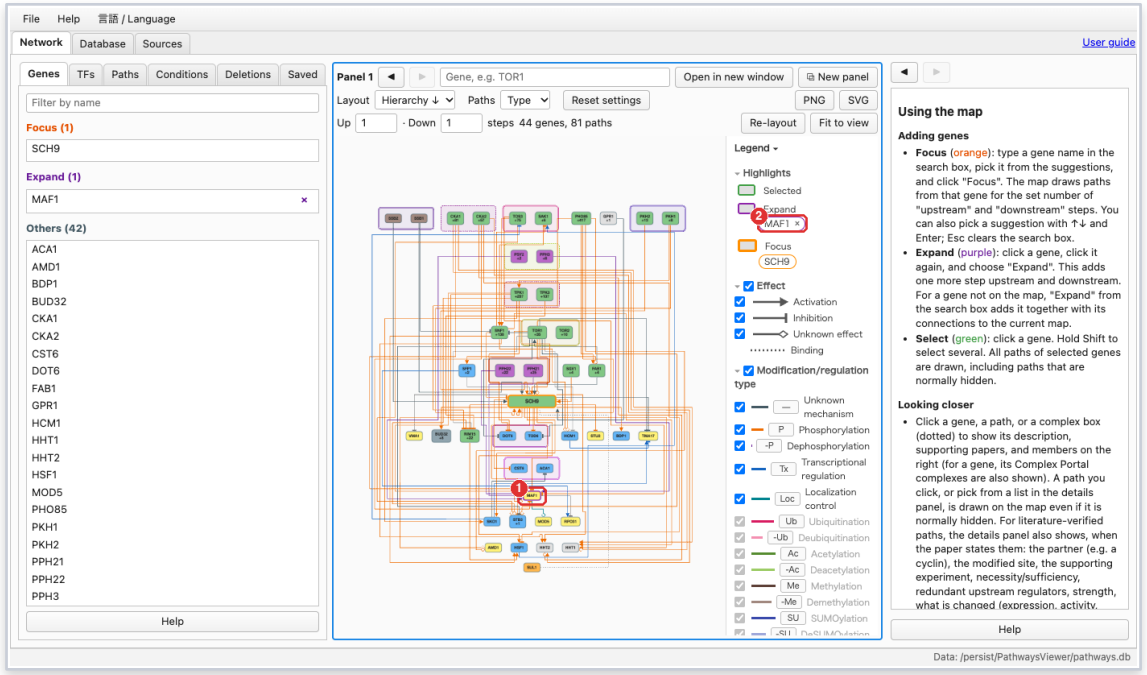
Map centred on SCH9

1. Enter a standard or systematic gene name in the search box ①, e.g. SCH9 or YHR205W.
2. Select Focus ②. The query gene is outlined in orange.
3. Set the upstream and downstream depth ②. A greater depth adds many genes and makes the map slower.

Selection and expansion



MAF1 selected

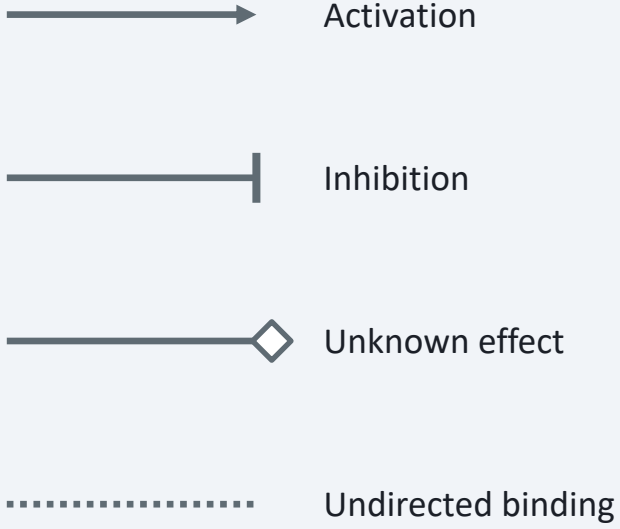


Map after expanding MAF1

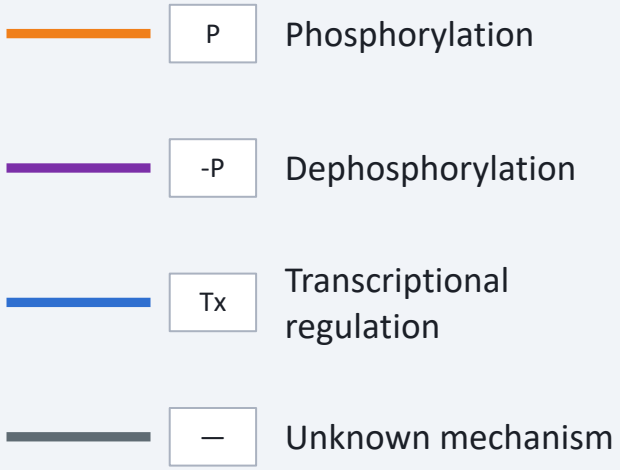
1. Clicking a gene selects it ①. Only its relations remain highlighted, and its annotation appears in the details panel ②. Shift-click selects several genes; clicking the background clears the selection.
2. Clicking a selected gene again offers Expand and Focus. Expand keeps the current map and adds one step upstream and downstream of that gene.
3. Expanded genes are outlined in purple ① and can be removed with × in the legend ②. Focus redraws the map around the gene.

Map notation

Edge terminus: effect

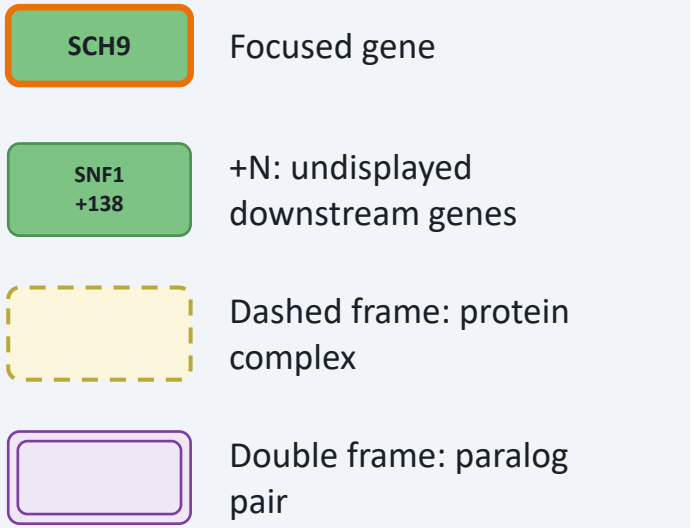


Edge colour: mechanism



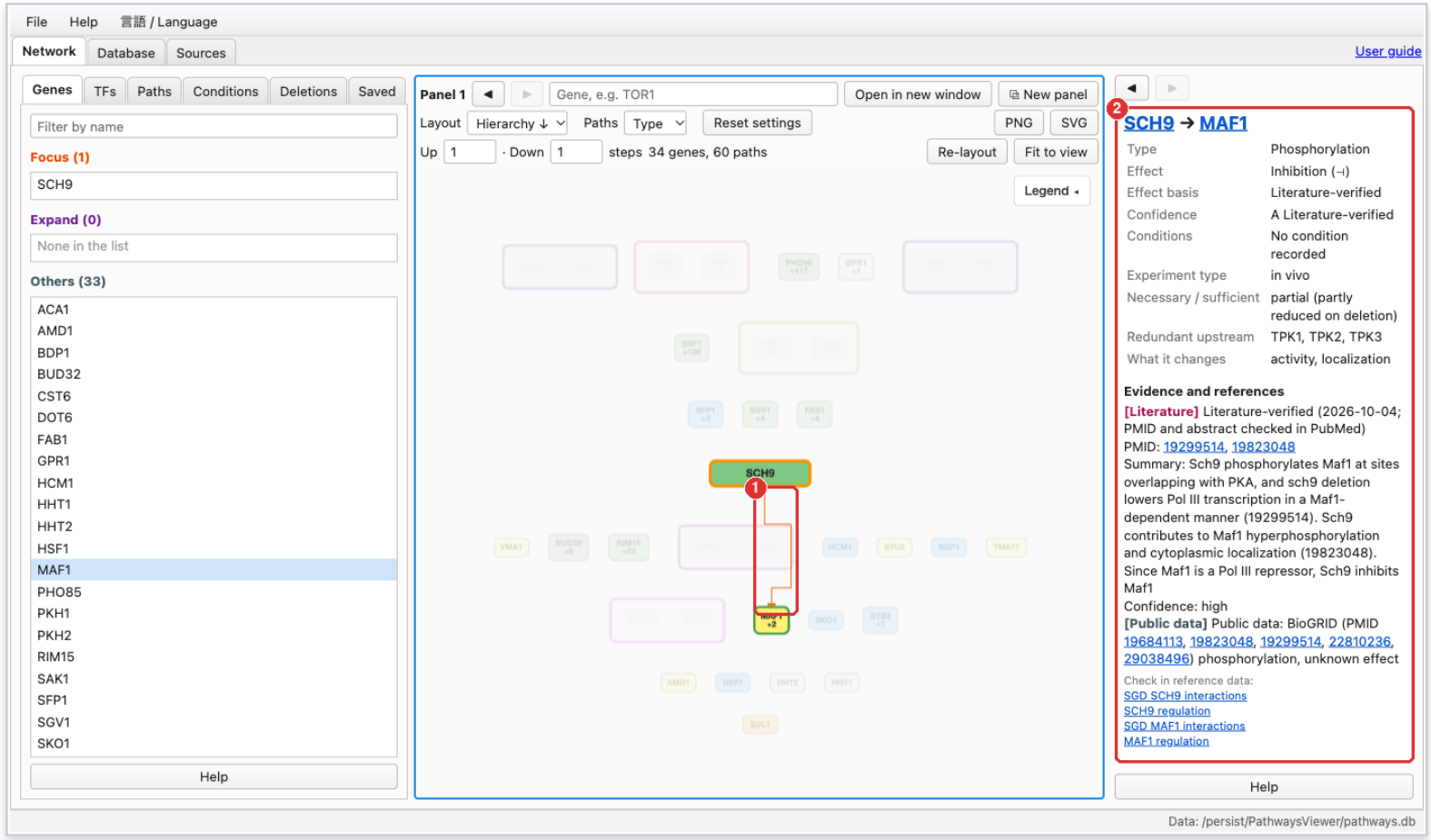
Edges can instead be coloured by effect: red for activation, blue for inhibition.

Nodes and frames



Upstream genes are placed higher. Node colour indicates functional class.

Evidence for each relation



1 **Selecting a relation**

Clicking an edge shows the relation in the details panel.

2 **Evidence panel**

Mechanism, effect, confidence grade A–D, conditions, PubMed identifiers and links to SGD and YEASTRACT+.

Left-hand tabs

Genes

Lists genes on the current map, grouped as focused, expanded and others. Selecting a gene highlights it and shows its annotation.

TFs

Lists transcription factors regulating the focused and expanded genes, grouped by effect. Ticked factors are added to the top row of the map.

Paths

Searches for paths between ticked genes with a specified number of steps and highlights the selected path.

Conditions

Filters relations by the conditions under which they were reported. Genes mode highlights genes whose expression or phosphorylation changes.

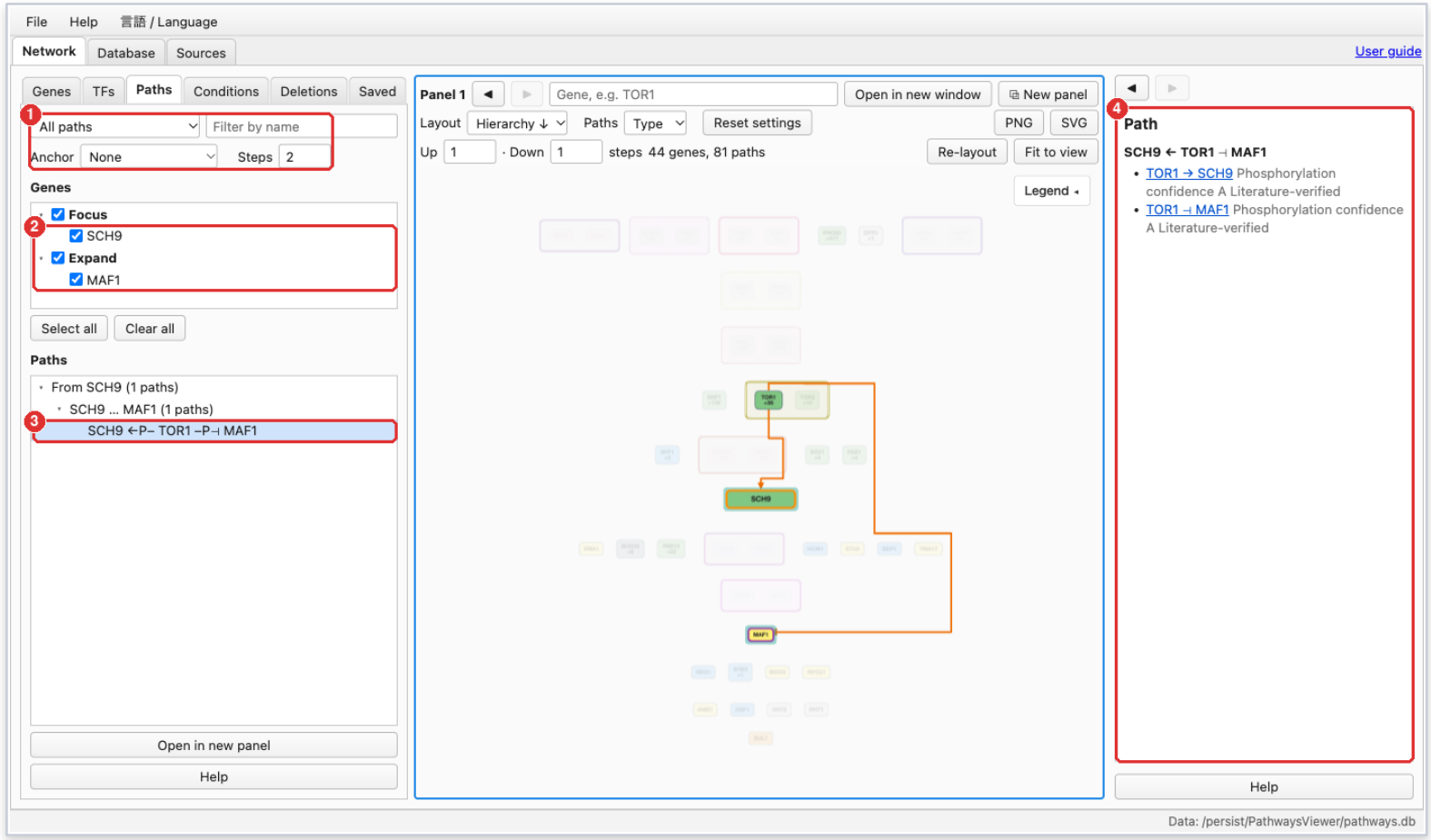
Deletions

Colours the map by mRNA or phosphorylation changes measured in deletion strains, and lists strains that affect a given gene.

Saved

Stores the current map state under a name for later recall. Saved views are kept in the browser.

Paths: connections between genes



1 Search mode

Select the path type and the number of steps.

2 Genes

Tick the focused or expanded genes to be connected.

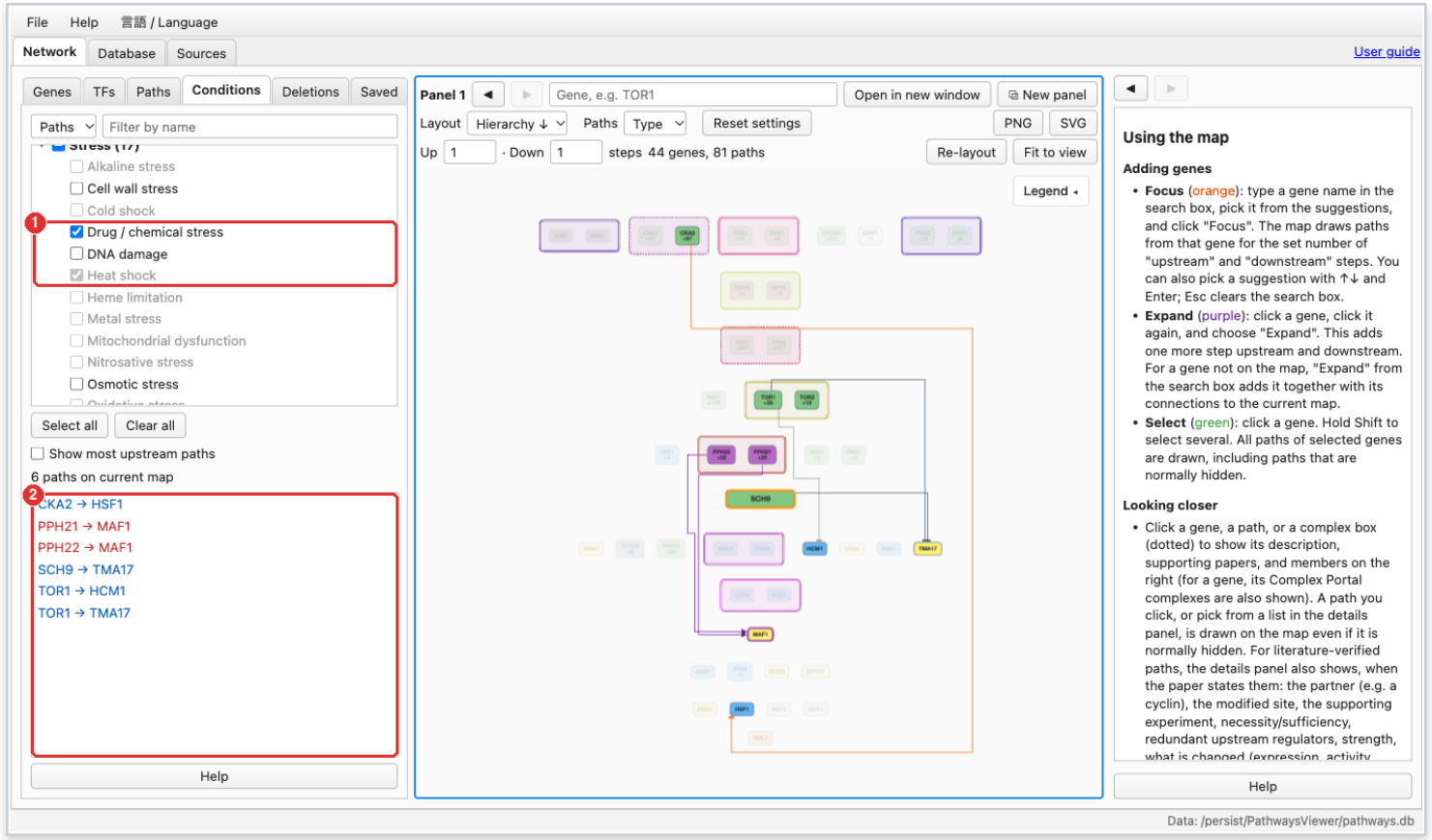
3 Results

Selecting a path highlights it on the map.

4 Path details

Each step is listed with its confidence grade.

Conditions: condition-specific relations



1 **Conditions**

Choose Paths or Genes at the top left. Clear all, then tick the conditions of interest, such as heat shock or drug stress.

2 **Matching relations**

Only relations reported under these conditions remain on the map.

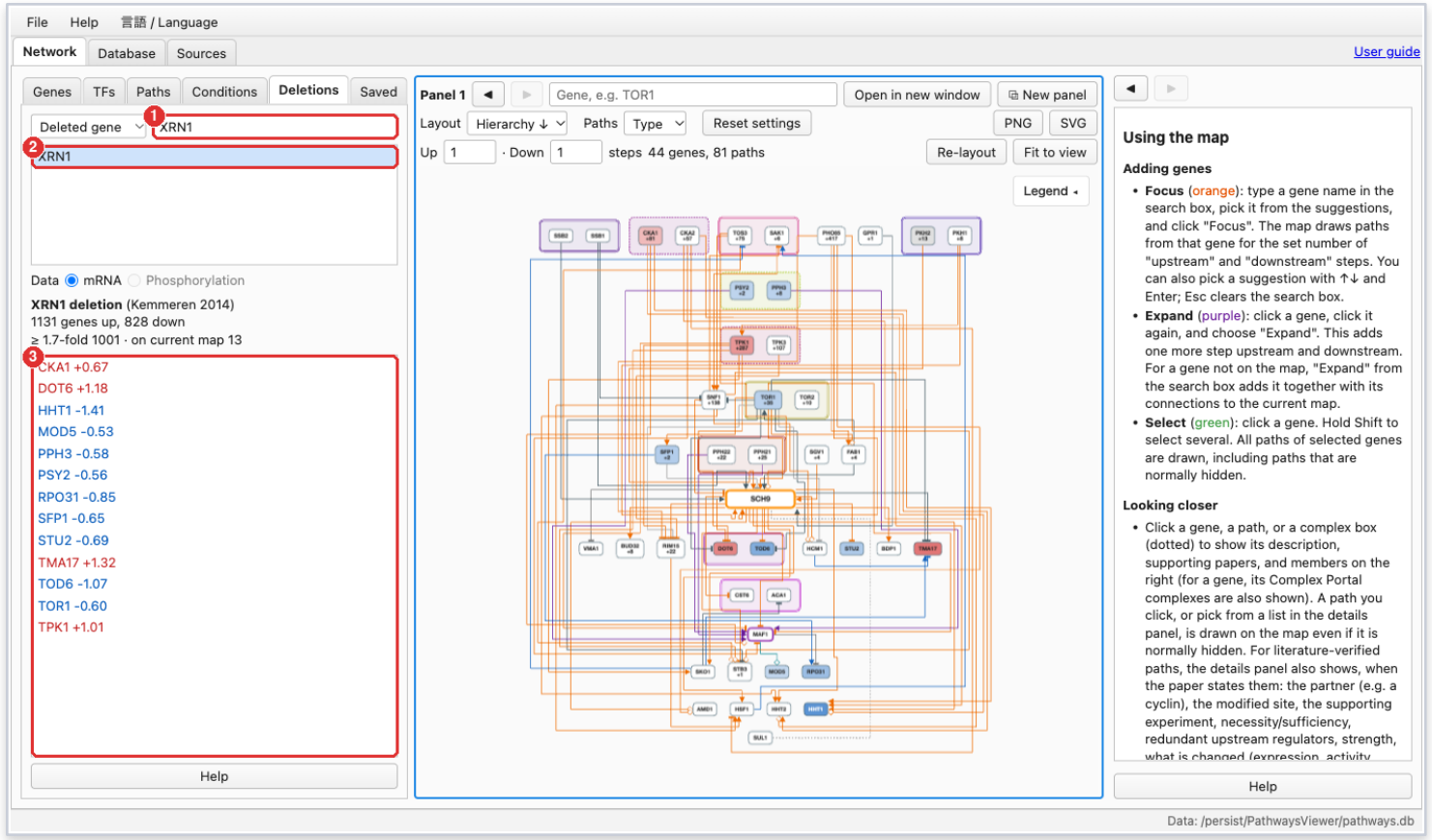
Paths

Relations reported under the selected conditions remain on the map.

Genes

Genes whose mRNA or phosphorylation changes at least twofold under the selected conditions are highlighted.

Deletions: measurements in deletion strains



1 Strain search

Choose Deleted gene or Changed gene at the top left, then enter a gene name, e.g. XRN1.

2 Strain selection

Nodes are coloured by measured change: red, increased; blue, decreased.

3 Changed genes

Genes on the current map that changed in the strain.

mRNA: Kemmeren et al. 2014, 1,481 strains. Phosphorylation: Bodenmiller et al. 2010, 113 kinase and phosphatase strains.

Deleted gene

Select a deletion strain; the map is coloured by the changes measured in that strain.

Changed gene

Select a gene on the map; the strains in which it changed are listed.

Additional functions

Multiple panels

Up to four maps can be compared side by side.

Database tab

Genes, relations and categories in tabular form, with filtering.

Export

Maps can be saved as PNG or SVG.

History

Previous maps and details can be revisited.

Help

Each panel has a Help button; the full user guide is linked at the top right.

Language

Japanese and English are available from the Language menu.